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Rehabilitation of human hearing with a totally implantable cochlear implant: a feasibility study

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Abstract

Background Cochlear implants (CIs) are neuroprosthetic devices which restore hearing in severe-to-profound hearing loss through electrical stimulation of the auditory nerve. Current CIs use an externally worn audio processor. A long-term goal in the field has been to develop a device in which all components are contained within a single implant. Here, we present initial clinical results with the totally implantable cochlear implant (TICI). The primary objective of this study was to assess the safety of the device in adults who suffer from bilateral severe-to-profound sensorineural hearing loss.

Methods This study used a design with non-randomized single group assignment (trial registration: NCT04571333). Six implantations took place beginning in September 2020. Data collection took place at the two participating CI centers. Adverse events (the primary outcome), speech perception, patient reported outcomes, and device usage statistics were collected over the subsequent 52 weeks. A within-subjects comparison was used in which each participant was evaluated both with the TICI and with an external SONNET audio processor.

Results One anticipated serious adverse device effect (ASADE) occurred. After treatment the event resolved without sequelae. No unanticipated serious adverse device effects (USADE) occurred. Speech perception in quiet and in noise scores were comparable between the TICI and the SONNET audio processor. Scores on the validated patient reported outcome instruments HUI3, SSQ-12, and HISQUI-19 all increased over the duration of the study. User satisfaction scores as reported in their daily diary also increased over the duration of the study. Based on device usage metrics, all but one user used the TICI without an external processor the majority of the time.

Conclusions The primary outcome of assessing the safety of the device was achieved. The TICI provides high levels of hearing performance, comparable to those of a conventional CI. The development of the TICI expands the range of options for treatment of hearing loss.

Plain language summary

Cochlear implants (CI) are used to treat severe-to-profound hearing loss. The current generation of CIs use an externally worn audio processor to capture sounds and convert them to electrical signals. These signals are then transmitted to an implant which electrically stimulates the cochlea to produce sound sensations. A totally implantable CI (TICI) has been developed in which the components of the audio processor are integrated into the implant, removing the need to wear an external audio processor. In this study, the authors present clinical results from a feasibility study where the TICI was implanted in six patients. Safety and performance with the TICI were comparable to a conventional CI. This device will benefit people with severe-to-profound hearing loss.

Hearing is an important sense, not just for social communication, but for perceiving and navigating the world around us. Severe-to-profound hearing loss (HL) is the near or total absence of this sense. Cochlear implants (CI) have been transformative in the treatment of severe-to-profound HL. Although there is still considerable variability in hearing outcomes among CI users, most users achieve substantial benefit, and some users achieve speech perception in quiet which is close to that of individuals with normal hearing.

Despite these benefits, CI uptake remains rather low. The WHO estimates that sixty million people worldwide have severe-to-profound HL[1], yet only slightly more than one million implantations, cumulatively, have taken place since the first operations several decades ago[2]. Several reasons for this

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Table 1 | Demographic and other baseline characteristics

ID	Age (decade)	Sex	Ear to be implanted	Etiology	HL type	Years since HL onset	Previous CI
1	20–30	Male	Right	Hereditary, genetic	Progressive	15	No
2	70–80	Female	Left	Genetic, progressive	Progressive	9	No
3	50–60	Female	Left	Hereditary, progressive	Progressive	32	Yes (right)
4	20–30	Female	Left	Meningitis, progressive	Progressive	18	Yes (right)
5	50–60	Male	Left	Hereditary, unknown	Unknown	NR	No
6	20–30	Female	Right	Unknown, sudden	Sudden	NR	Yes (left)

Age and years since the onset of hearing loss are calculated from the time of inclusion in the study. Age is given in decades to preserve participant anonymity.

low uptake have been identified[3]. One aspect is related to the audio processor. The audio processor is an externally worn device that detects speech and other sounds, encodes these as electrical signals, and transmits them to the implant. As an external device, an audio processor can be lost or damaged. It requires cleaning, maintenance, and additional peripherals to operate. It can interfere with activities such as bathing or swimming. It is also true that the wearing of a hearing prosthesis, a visible marker of disability, can be a source of cosmetic and self-esteem issues[4–8].

To solve this set of issues, we have developed and tested the totally implantable CI (TICI, device name: Mi2000). In this device, all components have been integrated into a single implant, which is invisible to the outside. This has required the resolution of numerous technical challenges, not least of which was the development of the subdermal microphone, the implantable power supply, and the transdermal wireless data link, all of which are required to be biocompatible and have operating lifetimes of decades.

The primary objective of this study was to assess the safety of the Mi2000 system in adults who suffer from bilateral severe to profound sensorineural hearing loss. The secondary objectives were to evaluate the initial clinical performance with the Mi2000 by assessing speech perception in quiet and in noise, as well as to assess user experiences with the device through several validated patient-reported outcome measures and a user satisfaction diary. The Mi2000 shows safety and performance characteristics comparable to that of a conventional CI. User self-rated hearing performance is good, as is their self-rated satisfaction and usage time.

Methods

Inclusion and exclusion criteria

The criteria for inclusion in this study were as follows: a minimum age of eighteen years at the time of enrollment; bilateral sensorineural hearing loss with no or limited benefit from a hearing aid (>70 dB HL PTA_4); postlingual onset of deafness; no or limited benefit from hearing aids for less than 10 years; a maximum score of 50% in the ear to be implanted on a monosyllable recognition test in the language of the study center; a general health condition and psychological and emotional condition deemed compatible with the treatment and tests performed in this study and realistic expectations, as deemed appropriate by the implanting surgeon/implant board; fluency in the test language with excellent proficiency, as appropriate to perform speech perception testing; evidence of up-to-date vaccinations against meningitis, as per recommendations by the applicable national body; and signed and dated informed consent before the start of any study-specific procedure.

The criteria for exclusion from the study were as follows: lack of compliance with any inclusion criterion; previously having received an implant on the location chosen for placing the TICI; having received a hearing implant from another manufacturer than MED-EL on the contralateral ear; having a pre-existing condition known to necessitate MRI scans after implantation of the Mi2000; women who were pregnant or nursing or of child-bearing age not reporting to use effective contraception; contraindication to surgery in the middle and inner ear; contraindication to general anesthesia; cochlear malformations, ossification or other obliteration of the cochlea; history of meningitis preventing placement of the electrode array, as confirmed by medical examination; acute cholesteatoma; acute external or middle ear infections; perforated tympanic membrane; known intolerance to any of the materials used for the implant or accessories; non-functional auditory nerve and/or upper auditory pathway including a history of vestibular schwannoma; factors preventing appropriate placement of the stimulator housing and the microphone, including fixation with screws, in the bone of the skull; unstable Ménière's disease, auditory neuropathy, or epilepsy not responding to treatment; unrealistic expectations of the participant; permanent inability and/or unwillingness to participation in post-operative care and rehabilitation; known intellectual disability and/or psychological diseases; and participation in other pharmacological clinical trials within four weeks prior to enrollment.

Participants, ethics, and trial registration

This study took place at the University Hospital Center of Liège, Belgium, and the University Hospital of Munich, Germany. Six participants (four female) were enrolled; three each at Liège and Munich. The first was enrolled on 16 September 2020, and the last on 16 December 2020. Demographic and other baseline characteristics are given in Table 1. The mean (±standard deviation, SD) age was 44.5 (±24.7) years. All had bilateral severe-to-profound HL (>70 dB HL measured by audiometry). The pre-operative audiometric pure tone averages (0.5, 1, 2, and 4 kHz) in the ear to be implanted ranged from 79 to 107 dB HL (mean ± SD: 9 ± 10 dB HL). One had sudden, and five had progressive HL. In the ear contralateral to the planned surgery site, three used an acoustic hearing aid, and three used a conventional CI. Two of the latter used a SONNET audio processor (MED-EL, Innsbruck, Austria), and one used a RONDO audio processor (MED-EL).

This study was designed and performed in accordance with the principles of the Declaration of Helsinki and with ISO 14155 (good clinical practice for clinical investigation of medical devices for human subjects). All participants gave their explicit written informed consent. This study was approved by the Comité d'Ethique Hospitalo-Facultaire Universitaire de Liège (Approval number: 2020/29) and the Ethikkommission bei der LMU München (Approval number: 20-161 fed). Safety was closely monitored throughout this clinical investigation. This trial was registered at clinicaltrials.gov (trial registration: NCT04571333).

Primary and secondary outcome measures

The primary outcome measure of this feasibility study was to describe the occurrence of adverse events (AEs) over the defined follow-up window, a summary of the number (*n*) and percentage (%) in terms of the number and percentage of participants with each type of AE.

The secondary outcome measures were to perform speech perception testing in quiet and in noise, to measure audiograms in the unaided and best-aided conditions, to measure participant self-reported quality of life and perceived sound quality outcomes via four validated instruments (detailed below); and to measure user satisfaction via a daily self-reported satisfaction rating diary. In addition to these, hardware and device parameters, including precision impedance field telemetry and auditory nerve response telemetry, were also recorded, and an *ad hoc* questionnaire was

Table 2 | Distribution of adverse events during the study

Event type	Number	Percent of AEs
Adverse event (AE)	5	33.3
Serious adverse event (SAE)	1	6.7
Adverse device effect (ADE)	8	53.3
Anticipated serious adverse device effect (ASADE)	1	6.7
Unanticipated serious adverse device effect (USADE)	0	0.0
Total	15	100.0

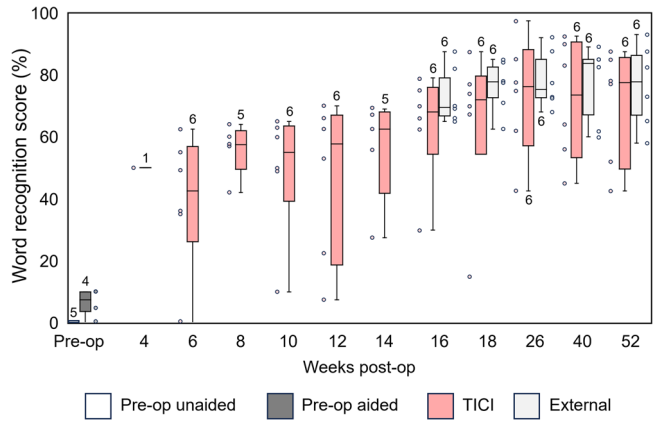


Fig. 1 | Monosyllabic word recognition scores at pre-operative and post-operative follow-up intervals. Word recognition scores were measured with the TICI at the 4- to 52-week post-operative intervals and with the external processor at the 16- to 52-week post-operative intervals. Individual scores are shown beside the box plots, in the corresponding color (left for the TICI, right for the external processor). The number of participants tested is shown above each box plot. Note that in this figure and the others, some of the points depicting individual scores may overlap. All five users who were tested pre-operatively in the unaided condition had word recognition scores of 0%. Error bars represent 1.5× the interquartile range.

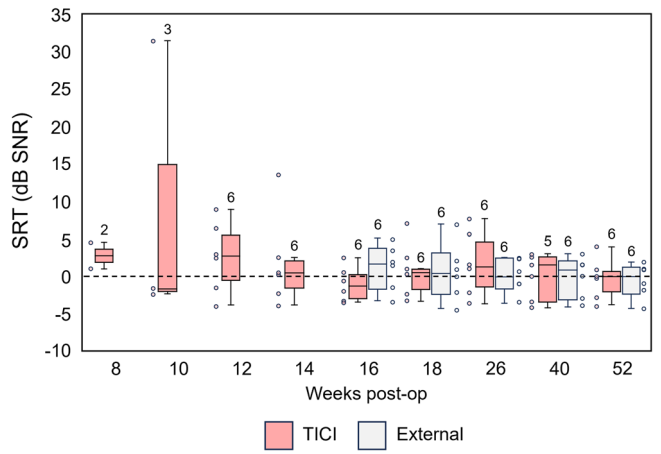


Fig. 2 | Speech reception thresholds in noise. Thresholds were measured with the TICI at the 8- to 52-week post-operative intervals and with the external processor at the 16- to 52-week post-operative intervals. Individual thresholds are shown beside the box plots, in the corresponding color (left for the TICI, right for the external processor). The number of participants tested is shown above each box plot. The dotted line indicates an SRT_{50} of 0 dB SNR (i.e., where the participant would achieve 50% word recognition when the signal and noise are presented at the same level). Lower values than this indicate that the signal can be presented at a lower level than the noise to achieve 50% word recognition (i.e., performance is better). Error bars represent 1.5× the interquartile range.

administered to the clinical investigators involved in the study. These latter measures are not reported here as they were intended as technical feedback to the device manufacturer.

Audiometry

Pure-tone audiometry was conducted according to standard procedures at each clinic. Audiometric thresholds were obtained in the 125–8000 Hz frequency range. Pre-operative thresholds were evaluated in the unaided ipsilateral ear. Post-operative aided thresholds were evaluated with the TICI (see Supplementary Fig. 2).

Surgical procedures and fitting

Surgery was performed via the standard surgical approach for cochlear implantation, with which the TICI is compatible. Four candidates were implanted on the left side, and two were implanted on the right. The electrode array was the 28 mm FLEX28 array in three cases and the 31.5 mm FLEXSOFT array in three cases. The microphone was fixed in a flattened bone bed with screws. The positioning of the TICI is shown in Supplementary Fig. 1. The surgical procedure had a mean duration of 211 min (minimum: 98 min, maximum 324 min). Post-operative recovery was uneventful. Adverse events (AEs) were documented during the study and are reported below.

Four weeks after implantation, initial activation was carried out. This involves telemetry to test the status of the implant and electrodes, and programming to fit the stimulation parameters to each user's configuration of hearing loss. Participants were also given a conventional audio processor (MED-EL SONNET), which allows the implant to operate as a conventional CI. This was done to allow within-ear comparisons of performance between the TICI and a conventional audio processor.

Speech perception in quiet testing

Speech perception in quiet was assessed using open-set monosyllabic speech testing. Participants at the Munich clinic were assessed using the Freiburg Monosyllabic Test and those at the Liège clinic were assessed using the Monosyllabic Test of Fournier. Both tests are phonemically balanced with respect to their source languages. The tests consist of 20 lists, each list containing 20 (Freiburg) or 10 (Fournier) words each. The tests were performed in a quiet room with a single audiological loudspeaker placed at head level. Speech stimuli were presented in the free field at 65 dB SPL. No competing noise was employed.

At each testing session, the participant was presented with two lists and asked to repeat the words they had heard. The percentage of correctly repeated words was recorded. Before the testing session, each participant was allowed to perform one practice list to acclimatize to the experimental setup. Contralateral hearing aids and CIs were switched off for the test. If the participant had residual hearing in the contralateral ear, the clinic's standard masking procedures were applied. No feedback was provided to the participants regarding the correctness of the responses.

Speech perception in noise testing

To assess speech perception in noise, the signal-to-noise ratio was measured at which listeners achieved 50% word recognition in sentences presented in competing background noise (50% speech reception threshold, SRT_{50}). Participants at the Liège clinic were assessed with the French Matrix Test; those at the Munich clinic were assessed using the Oldenburger sentence test (OLSA).

The mean SRT of two trials, using 20 sentences each, was reported. Contralateral hearing aids and CIs were switched off for the test. If the participant had residual hearing in the contralateral ear, the clinic's standard masking procedures were applied. Before the test, the participants were allowed to perform practice lists in order to acclimatize to the procedure.

Patient-reported outcome measures

Four widely used instruments for patient-reported outcomes were used in this study. These were the Health Utilities Index mark 3 (HUI3)[9], the 12-

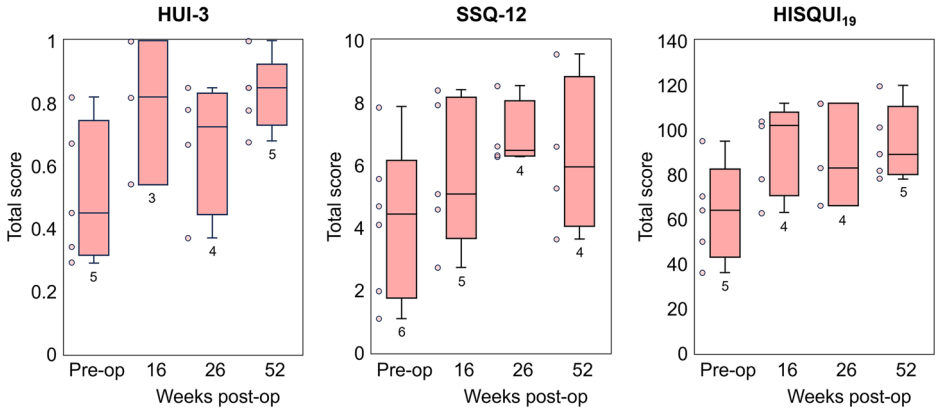


Fig. 3 | Patient-reported outcome measures on the HUI3, SSQ-12, and HISQUI-19 instruments. For each instrument, total scores at the pre-operative and the 16-, 26-, and 52-week post-operative intervals are shown. Individual scores are shown to the left of each box plot. The number of participants tested is shown below each box plot. Error bars represent 1.5× the interquartile range.

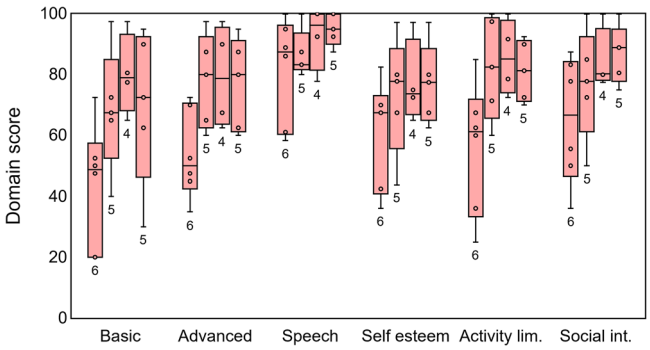


Fig. 4 | Patient-reported outcome measures on the NCIQ-60. Separate scores are shown for each of the six domains. Within each domain, the four box plots represent pre-operative and 16-, 26- and 52-week post-operative measurements, respectively. Individual scores are given within each box plot. The number of participants tested is shown below each box plot. Scores can range from 0 to 100. Basic: basic sound perception, Advanced: advanced sound perception, Speech: speech perception, esteem: self-esteem, activities: activity limitations, social: social interaction. Error bars represent 1.5× the interquartile range.

item form of the Speech, Spatial and Qualities of Hearing scale (SSQ-12)[10], the Hearing Implant Sound Quality Index (HISQUI-19)[11], and the Nijmegen Cochlear Implant Questionnaire (NCIQ)[12].

User diary report

During the post-treatment visit, the participants were provided with a satisfaction diary and instructed on how to use it during the follow-up phase. They were asked to rate their satisfaction with the usage of the TICI on a daily basis. The participants were instructed to bring the diary with them to the study site at each follow-up visit.

Device usage statistics

Device usage metrics are stored on the on-board memory of the TICI and the SONNET audio processor. At each follow-up visit, these metrics were extracted using the MAESTRO software. The total device usage times and charging times over the study were normalized to the total participation time of each participant (i.e., the exact time between first fitting and v52). Data are presented as hours per day.

Statistics and reproducibility

Due to the small sample size, inferential statistics were not performed. Results are summarized descriptively. Quantitative data were presented as mean with standard deviation and/or median with range (minimum and maximum). Qualitative data were presented as absolute and relative frequencies. Only prespecified primary and secondary outcomes are reported. No post hoc exploratory were performed.

Results

Adverse events

Fifteen adverse events occurred during the study. The distributions of adverse events are given in Table 2. Nine adverse events were classed as adverse device effects (ADE), one of which was an anticipated serious adverse device effect (ASADE). This was swelling at the site of implantation, with signs of an infection. After treatment the event resolved without sequelae. No unanticipated serious adverse device effects (USADE) occurred during the study.

Speech perception in quiet and noise

Monosyllabic word recognition was evaluated pre-operatively in both aided and unaided conditions and post-operatively with the TICI at the 4- to 52-week follow-up intervals (Fig. 1). Participants were also evaluated using the external audio processor at the 16- to 52-week follow-up intervals.

Due to the difficulty of the test, speech perception in noise was not evaluated pre-operatively, nor at first fitting or the 6-week interval. Post-operatively, speech perception in noise was tested with the TICI at the 8- to 52-week follow-up intervals (Fig. 2). Participants were also evaluated using the external audio processor at the 16- to 52-week follow-up intervals.

Patient-reported outcome measures

These instruments were administered pre-operatively and at the 16-, 26-, and 52-week follow-up intervals. Total scores on the HUI3, SSQ-12, and HISQUI-19 instruments are given in Fig. 3. Total scores on the HUI3 can range from −0.36 to 1.00. From the pre-operative interval to the final follow-up interval, the median (IQR) total score increased from 0.45 (0.31–0.74) to 0.85 (0.73–0.92) (an increase of +0.40). Total scores on the SSQ-12 can range from 0 to 10. From the pre-operative interval to the final follow-up interval, the median (IQR) total score increased from 4.44 (1.75–6.15) to 5.95 (4.04–8.83) (an increase of +1.51). Total scores on the HISQUI-19 can range from 19 to 133. From the pre-operative interval to the final follow-up interval, the median (IQR) total score increased from 64 (43–82.5) to 89 (80–110.5) (an increase of +25).

Patient-reported outcome measures with the NCIQ-60 instrument are given in Fig. 4. This instrument does not have a standard method for calculating a total score, so scores on the six domains are presented separately. Scores on each domain can range from 0 to 100.

From the pre-operative interval to the final follow-up interval, the median (IQR) domain score increased from 48.8 (20.0–57.5) to 72.5 (43.6–92.5) (in increase of +23.7) for *basic sound perception*; from 50.0 (42.5–70.6) to 80.0 (60.0–91.3) (in increase of +30.0) for *advanced sound perception*; from 87.5 (60.4–96.3) to 95.0 (90.0–100.0) (in increase of +7.5) for *speech perception*; from 67.5 (40.9–73.1) to 77.5 (65.0–88.6) (in increase of +10.0) for *self-esteem*; from 61.3 (33.9–71.9) to 81.3 (71.3–91.3) (in increase of +20.0) for *activity limitations*; and from 66.7 (46.5–84.4) to 88.9 (77.8–95.0) (in increase of +22.2) for *social interaction*.

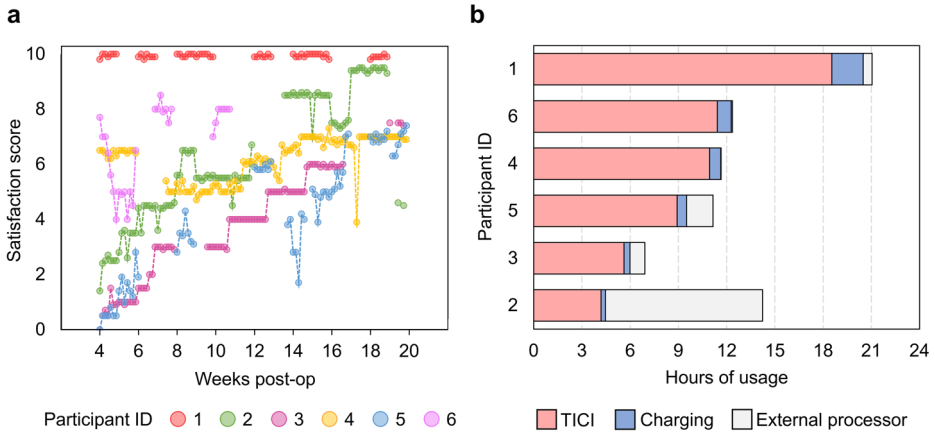


Fig. 5 | Participant satisfaction and device usage statistics. a Self-reported satisfaction scores at different post-operative intervals. Each point represents a single daily satisfaction report. Contiguous reports are connected with dotted lines. Non-contiguous reports are not connected. Reports are color-coded by the participant. **b** Device usage statistics in hours per day, averaged over the entire study period. Participants are sorted by TICI usage hours.

User satisfaction and device usage

User diary satisfaction scores and device usage statistics are presented in Fig. 5. Satisfaction scores can range from 0 (the lowest degree of satisfaction) to 10 (the highest degree). Each daily report from each user is shown. It can be seen that not all users provided reports on all days. Nevertheless, a trend of generally increasing satisfaction scores is evident.

Daily usage of the TICI (without an external audio processor) ranged from 4.2 h (participant 2) to 18.5 h (participant 4). Daily usage of the external audio processor ranged from 0 h (participant 4) to 9.8 h (participant 2). All but one user (participant 6) used the TICI without the external processor the majority of the time.

Discussion

Here, as far as we are aware, we have reported the first-in-human results with a novel totally implantable cochlear implant. The main objective of this feasibility study was to investigate the safety of the TICI. It was found that fifteen adverse events occurred during the study, but only one of these was a serious adverse device effect which, after appropriate treatment, resolved without sequelae. The distributions of adverse events were typical for those seen during conventional CI implantation. Overall, these results suggest that the safety of the TICI is comparable with that of a conventional CI.

For the objective speech perception tests, it was observed that the use of the TICI with its internal microphone and signal processor achieved similar scores as with the use of the external processor. Results with the external processor were quite typical of those of users implanted with a regular CI. These findings suggest that the implanted components of the TICI achieve similar clinical performance outcomes as those with a conventional CI.

For the validated patient-reported outcome instruments (HUI-3, SSQ-12, HISQUI19, and NCIQ-60), a generally increasing trend was observed from the pre-operative interval to the final post-operative interval, and high scores were achieved overall. These findings suggest that the participants' own perceived benefit from using the device was high.

For user satisfaction, a trend of generally increasing scores over time was observed. From the device usage statistics, it was found that five of the six participants used the TICI without an external processor considerably more frequently than with an external processor. There was a single outlier to this trend (ID 2). It is not known exactly why this participant preferred to use the external processor, but during a follow-up visit, this participant noted that they have a pattern of use in which they tend to use the device with the external processor while at home and use it without the device while out in public.

In summary, these results show that the TICI can be implanted via the conventional surgical approach. The data suggest that the TICI can yield hearing performance comparable to that of a conventional CI and can provide substantially improved hearing and quality-of-life outcomes for users. These conclusions can only be tentative at present—due to the small sample size, we were unable to conduct robust inferential statistics to assess the outcomes reported here. This was a first-in-human trial of an active neuroprosthetic device which is implanted via an invasive surgery. Such trials, by their nature, must be small.

Previously, a pilot study was conducted with another experimental fully implantable CI[13]. Our clinical results compare favorably with those. With the previous device, test performance was considerably poorer when using the internal device alone compared to using the external processor. In contrast, the Mi2000 TICI delivers within-user speech perception outcomes comparable to those with a conventional external audio processor.

The successful development of the totally implantable CI can be regarded as a milestone in the field. CIs are by far the most widely used neural prosthetic devices today. The lessons learned here may help to advance the field in general, and will inform a future pivotal trial to assess the efficacy of the TICI.

Data availability

The data for this study are available in the supplementary data file, and the descriptions for these data are found in the supplementary document.

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Author contributions

P.L.: conception, acquisition, interpretation of data, critical revision, final approval, accountability. J.M.: conception, acquisition, interpretation of data, critical revision, final approval, accountability. G.M.: conception, analysis, interpretation of data, critical revision, final approval, accountability. F.S.: conception, acquisition, curation, analysis, interpretation of data, critical revision, final approval, accountability. I.H.: conception, funding, supervision, critical revision, final approval, accountability, K. Koinig (MED-EL) performed statistical analysis.

Competing interests

The authors declare the following competing interests: G.M., F.S., and I.H. are employees of MED-EL. MED-EL provided both the funding and the implants used in this study. MED-EL was also involved in the study design, data analysis, and paper preparation. P.L. and J.M. declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43856-024-00719-0.

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